Yanlu Wang

Enh. MR Fysik, Medicinsk Strålningsfysik och Nuklear Medicin

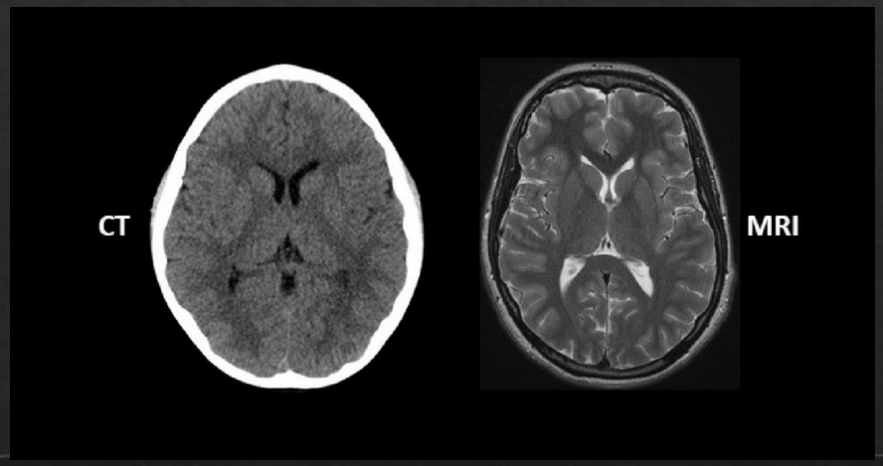
Karolinska Universitetssjukhuset, Solna

Tillämpningar MRT

Magnet Resonans Tomografi (MRT)

Fördelar

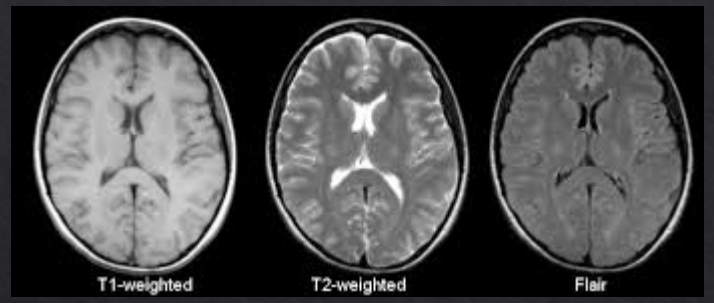
- Inga joniserande strålning
- Överlägsen mjukvävnadskontrast
- Många olika typer av kontrast kan avbildas

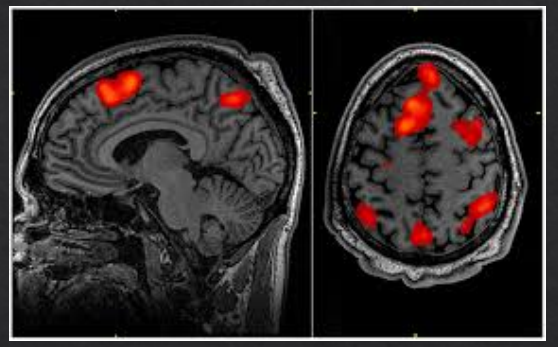


Nackdelar

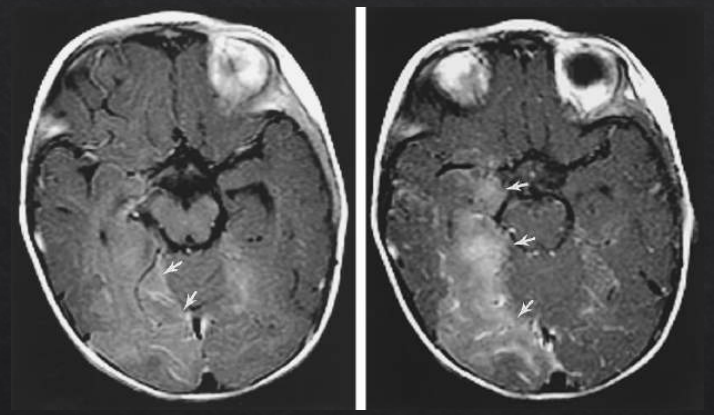
- Långsam
- Dyr
- Säkerhet och logistik
  - Inga (ferro)magnetiska föremål kan skannas
  - Implantat och pacemakers
- Relativa Intensitet/Kontrast
  - Svår att få absoluta värden med MR

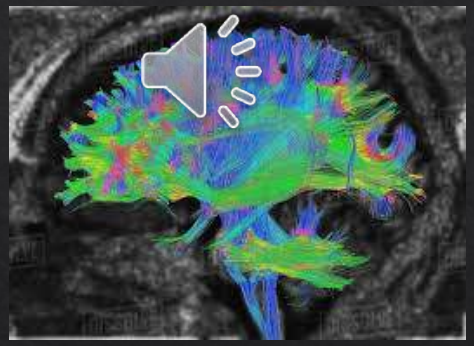
MR Neuro

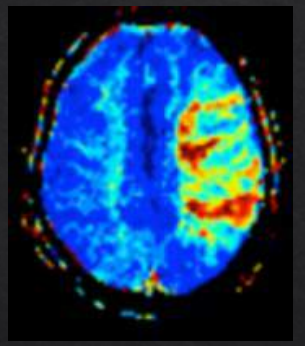




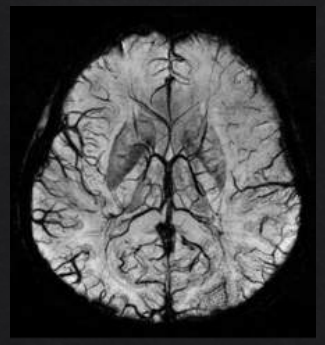


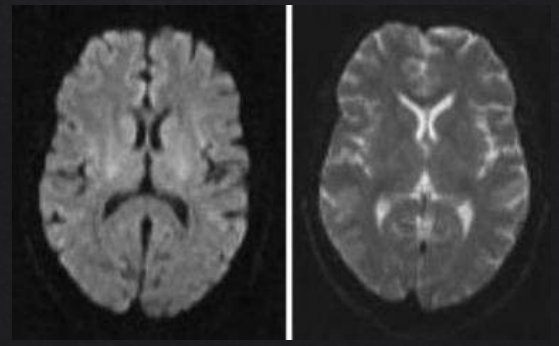








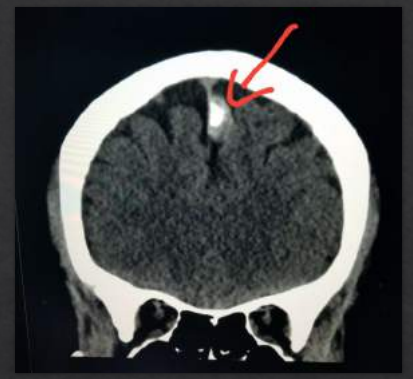


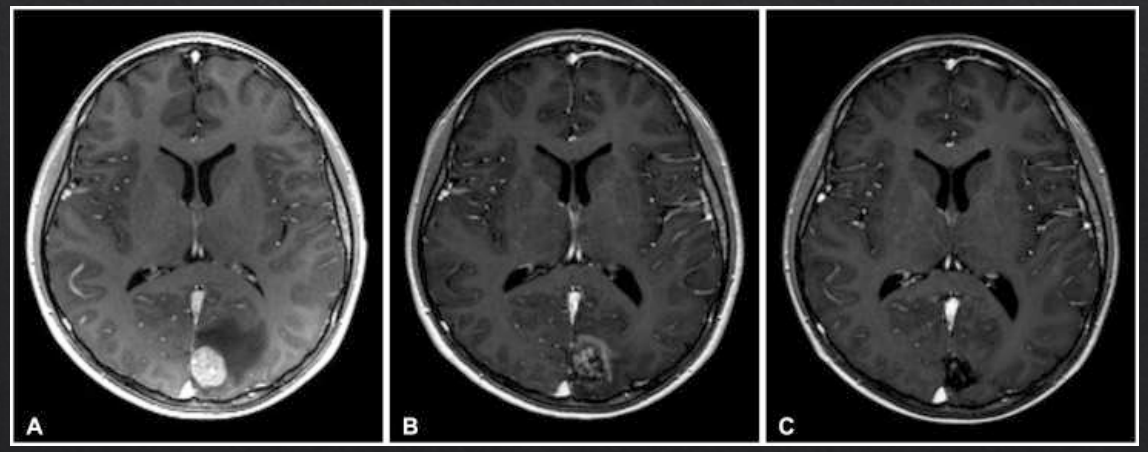


Tumör (1)

Bättre och manga olika typer av mjukvävnadskontrast:

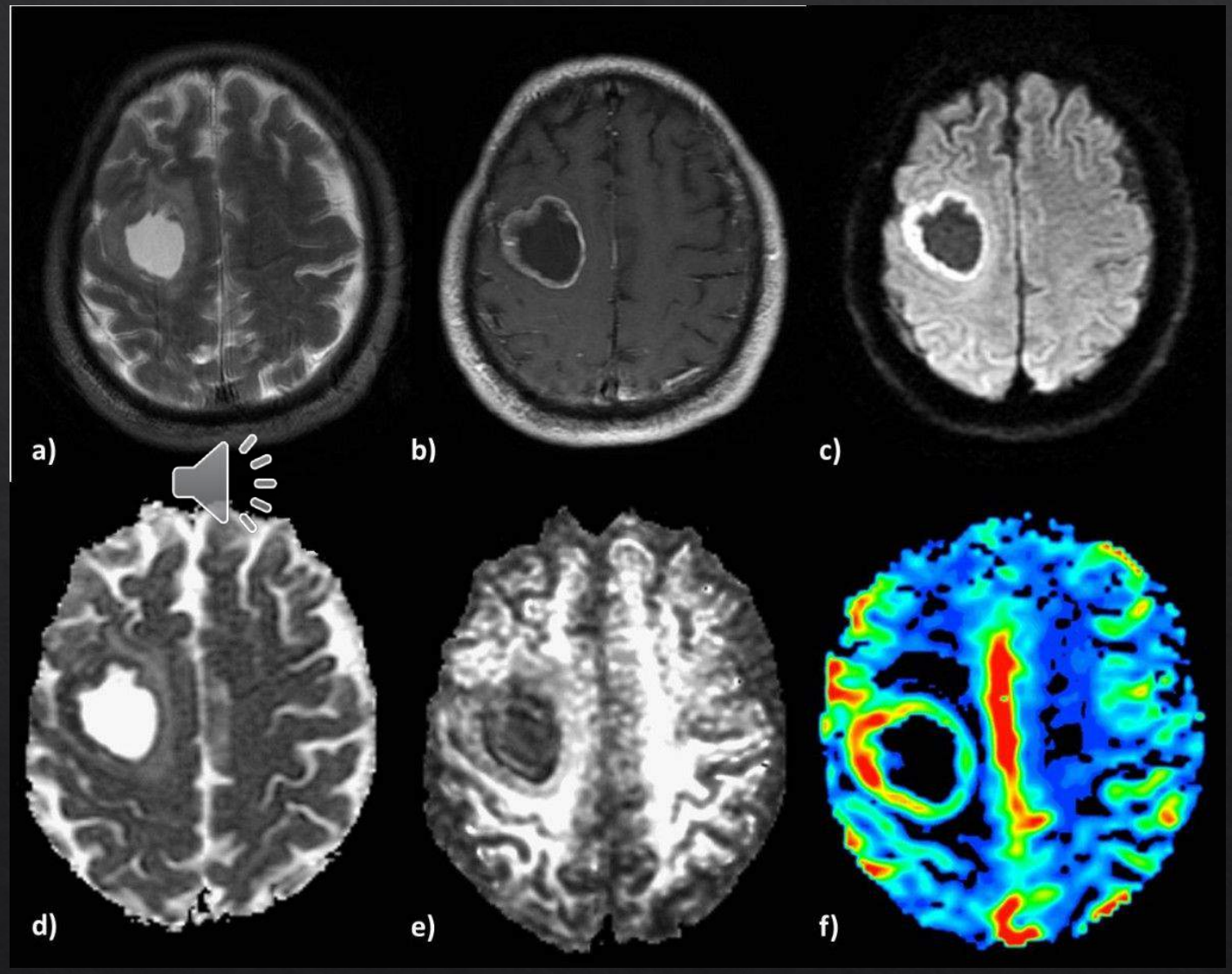
- Bättre särskilning av normal- och tumörvävnad.
- Kan enkelare skilja mellan nekrotisk vävnad, tumörvävnad, edem, blödning, etc…





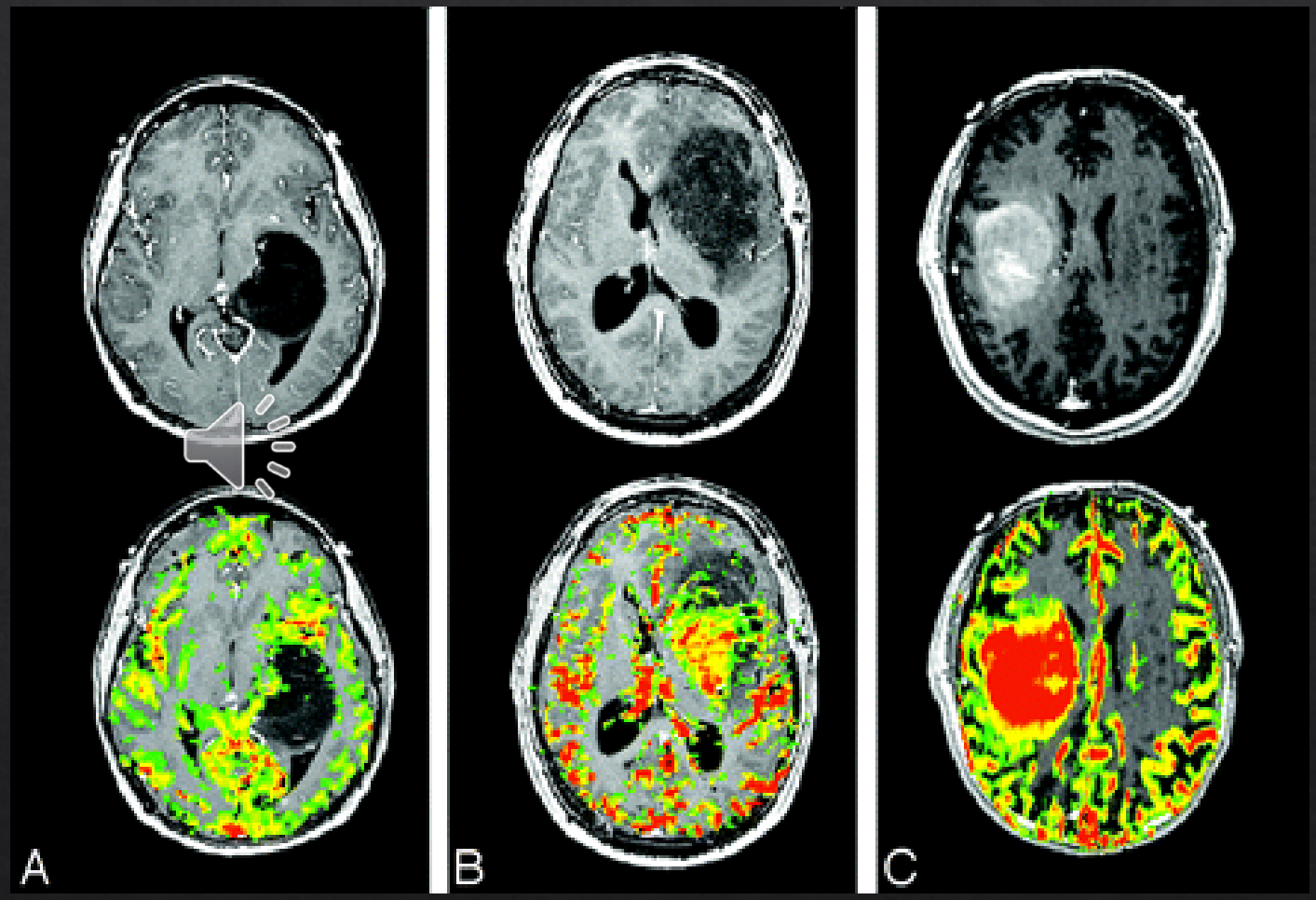
Tumör (2)

- ”Exotiska” kontraster som diffusionsviktning och perfusionsvikning kan också användas för att gradera tumörer.



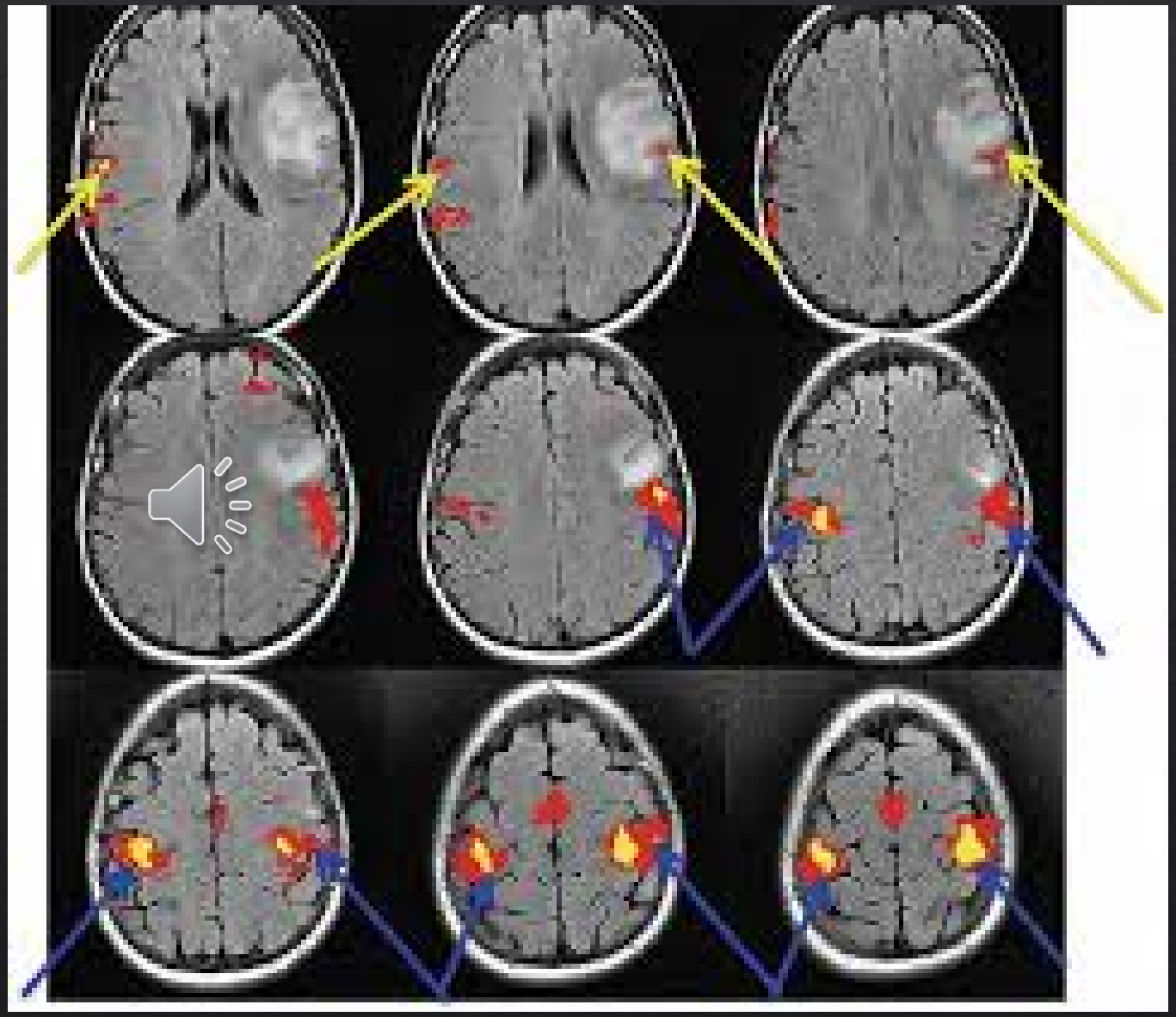
Tumör (3)

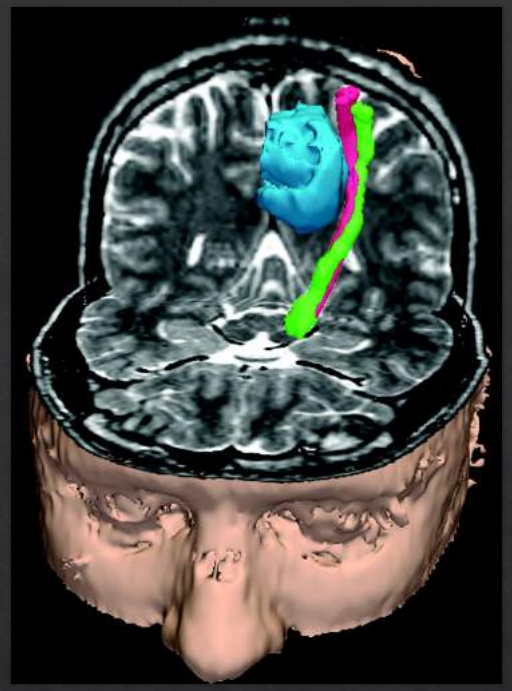
Staging

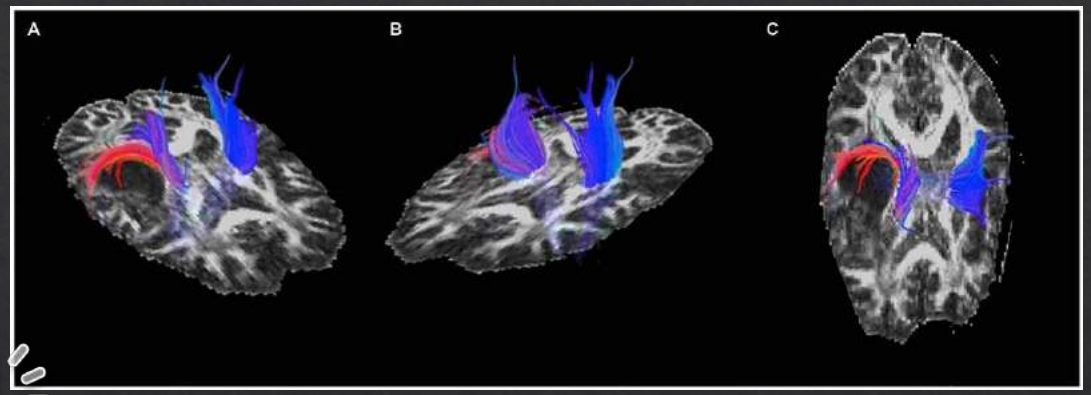


Tumör (4)

- Prekirurgisk planering (1)
  - Undvik funktionella delar med fMRI

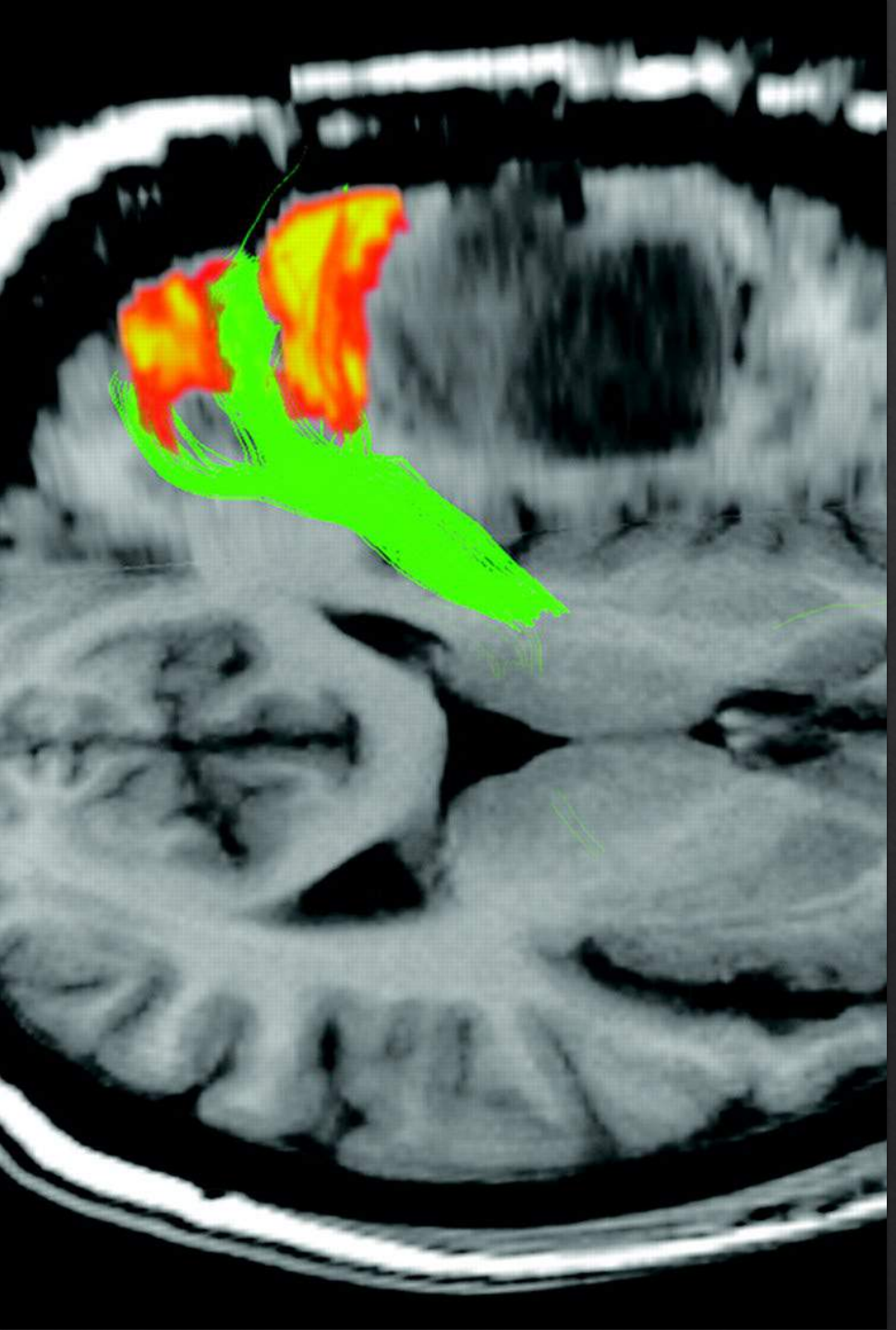






Tumör (5)

Undvik skada till “informationstransporten” med DTI

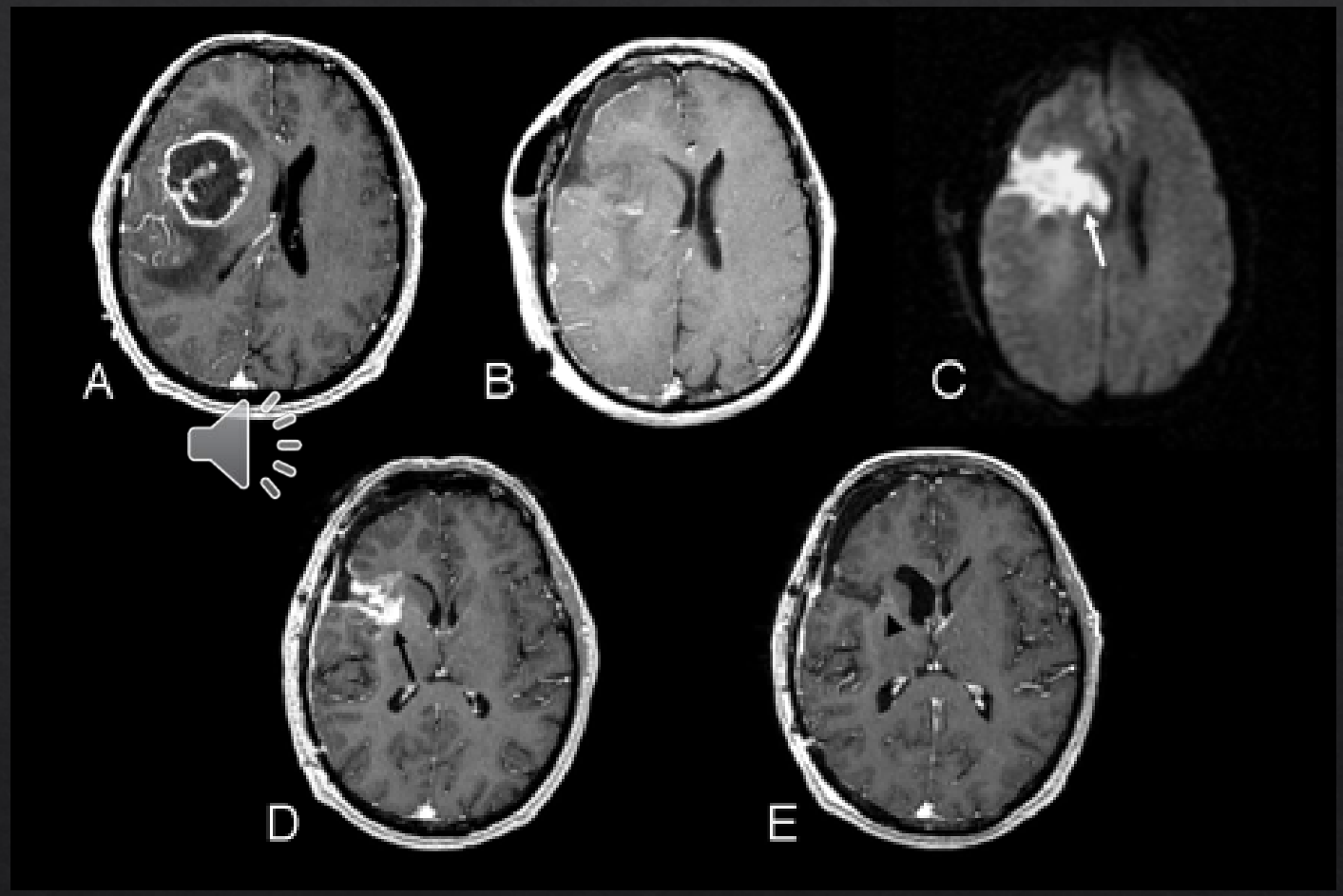


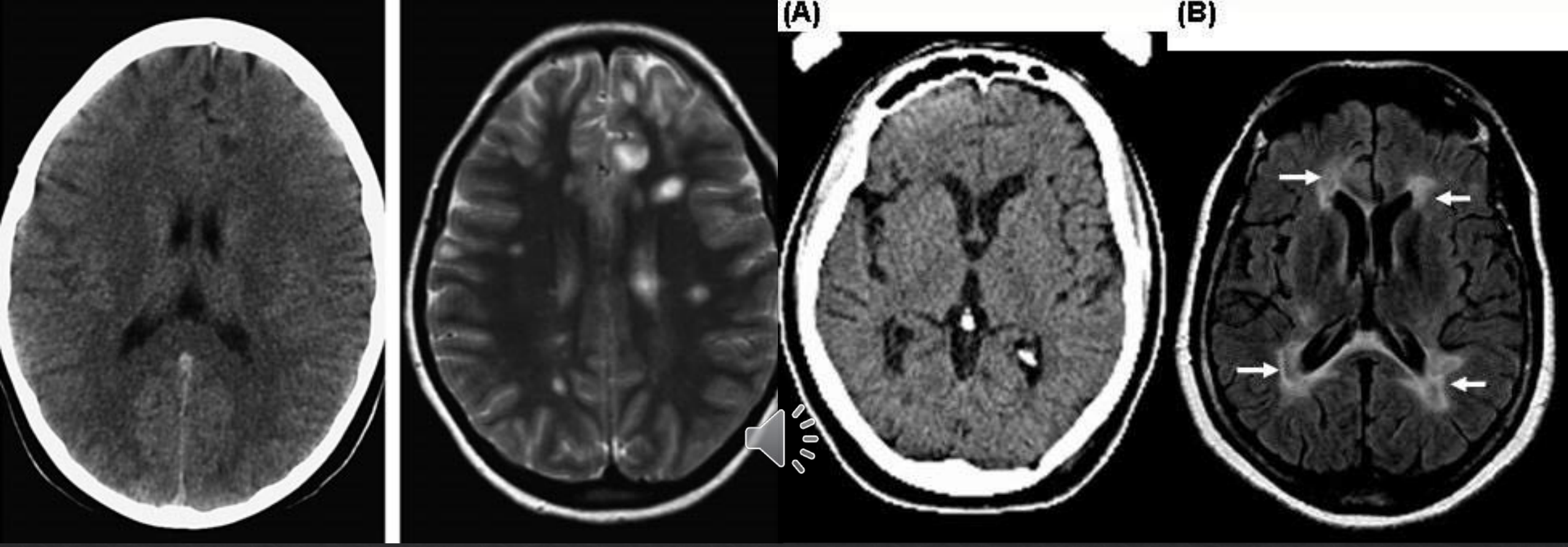
Tumör (6)

Kombinera fMRI och DTI

Tumor (5)

- Uppföljning

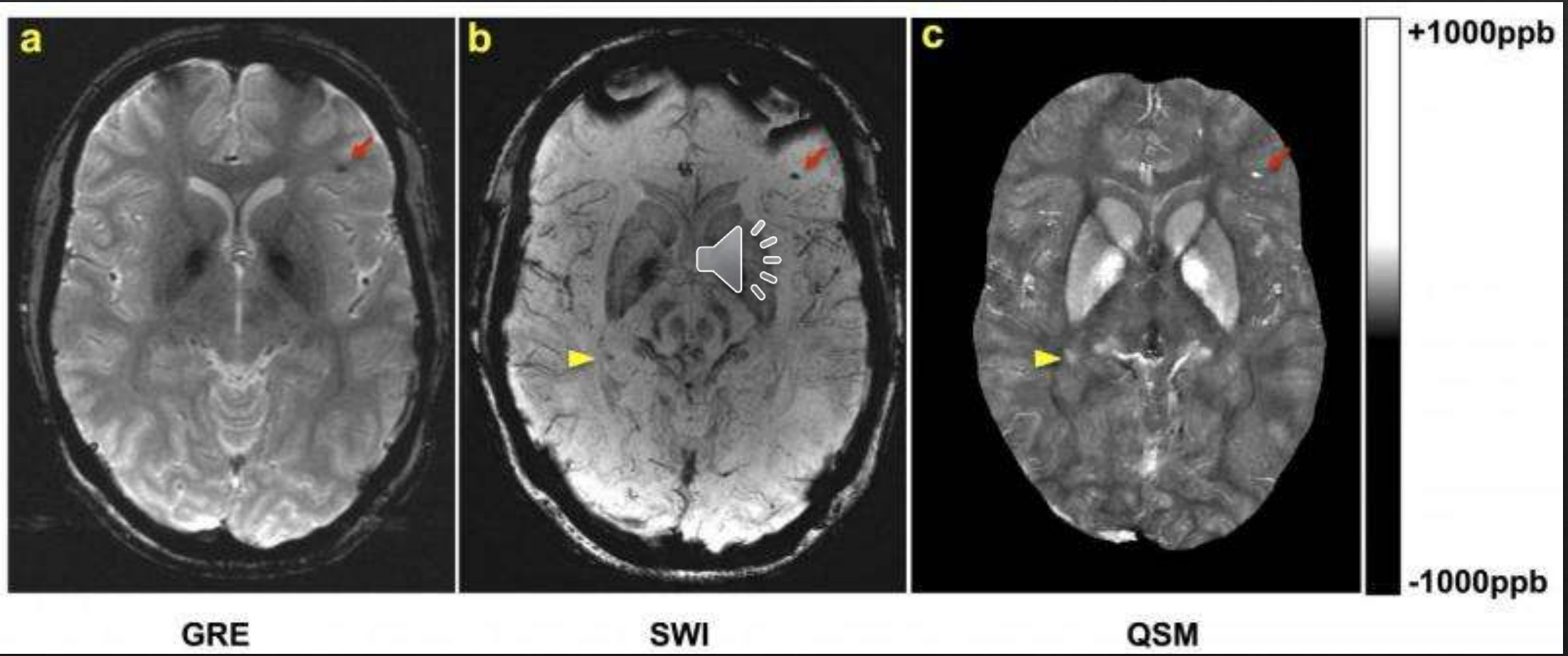




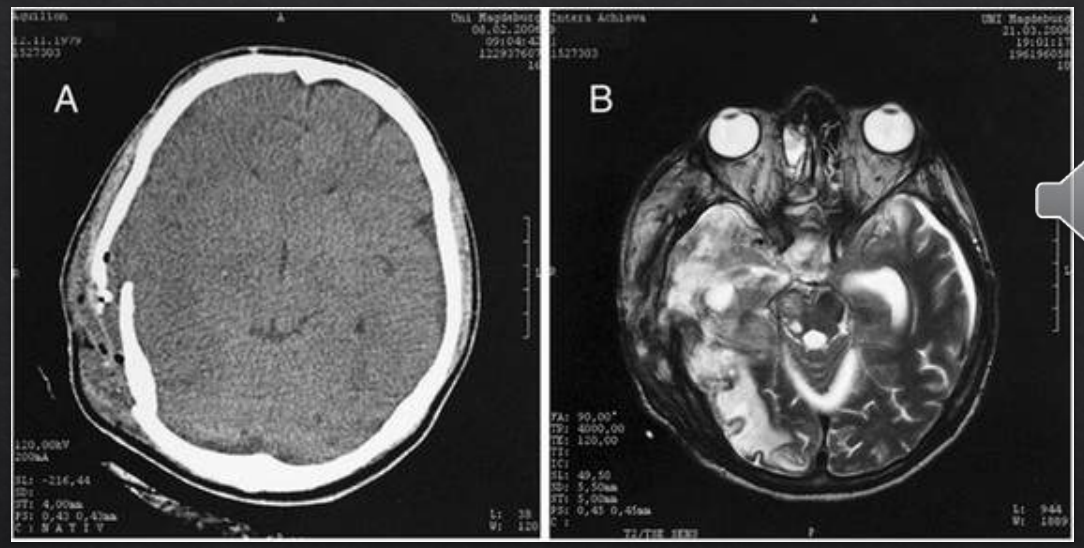
Neurologiska sjukdomar (1)

Multipel skleros

Neurologiska sjukdomar (2)



Trauma



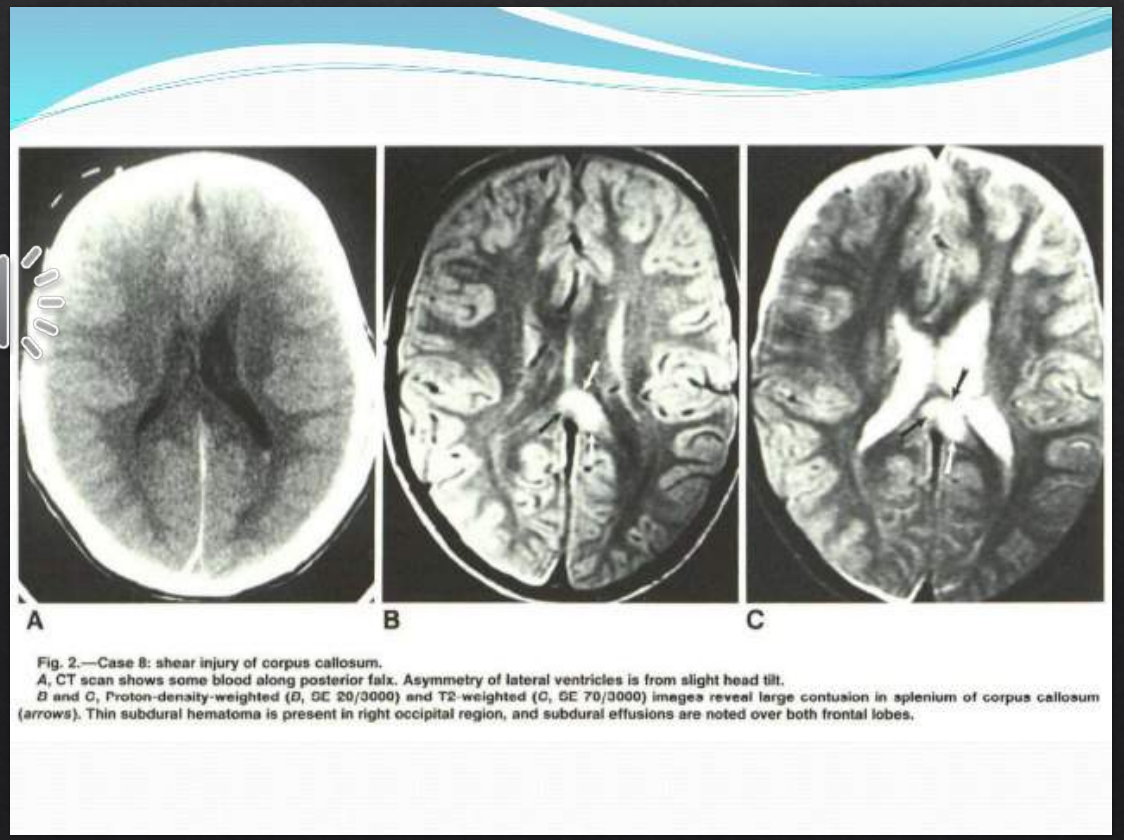
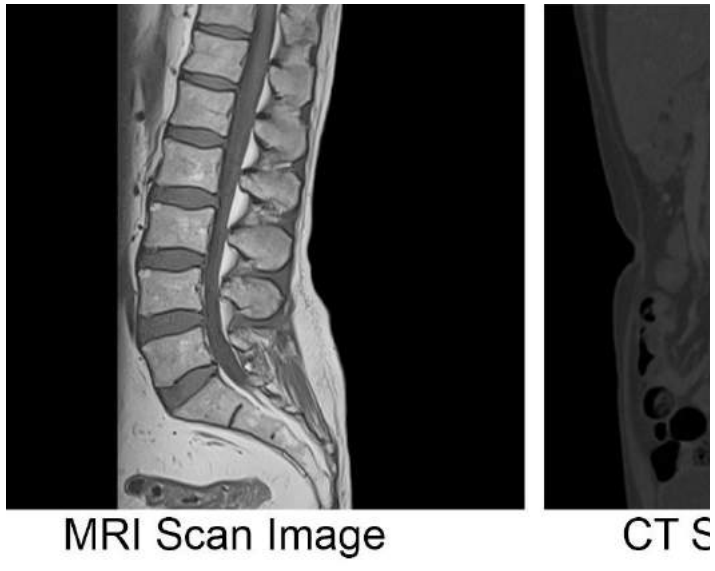


Fig. 2.—Case 8: shear injury of corpus callosum.
A, CT scan shows some blood along posterior falx. Asymmetry of lateral ventricles is from slight head tilt.
B and *C*, Proton-density-weighted (*B*, SE 20/3000) and T2-weighted (*C*, SE 70/3000) images reveal large contusion in splenium of corpus callosum (*arrows*). Thin subdural hematoma is present in right occipital region, and subdural effusions are noted over both frontal lobes.





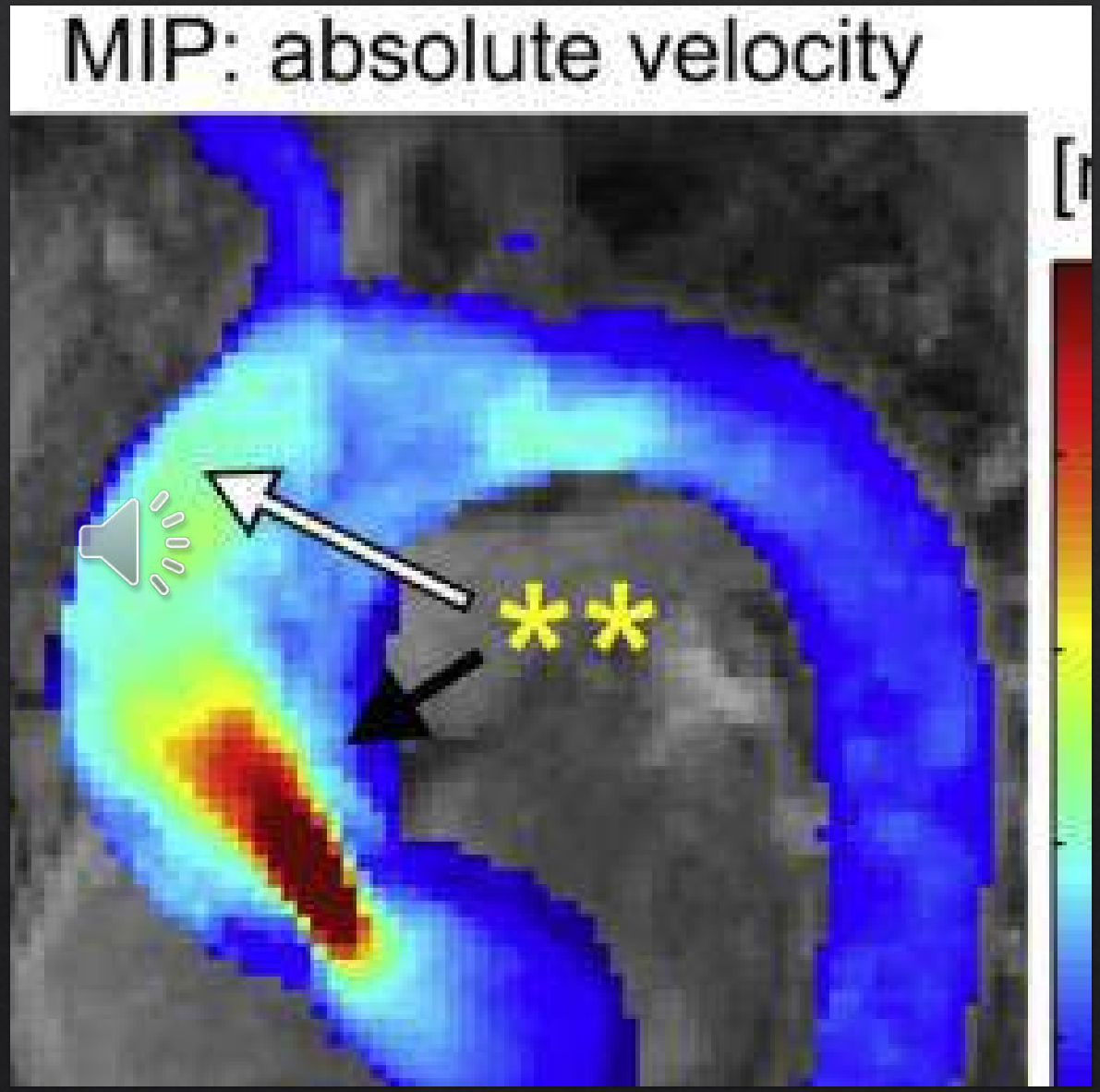


Diskbråck

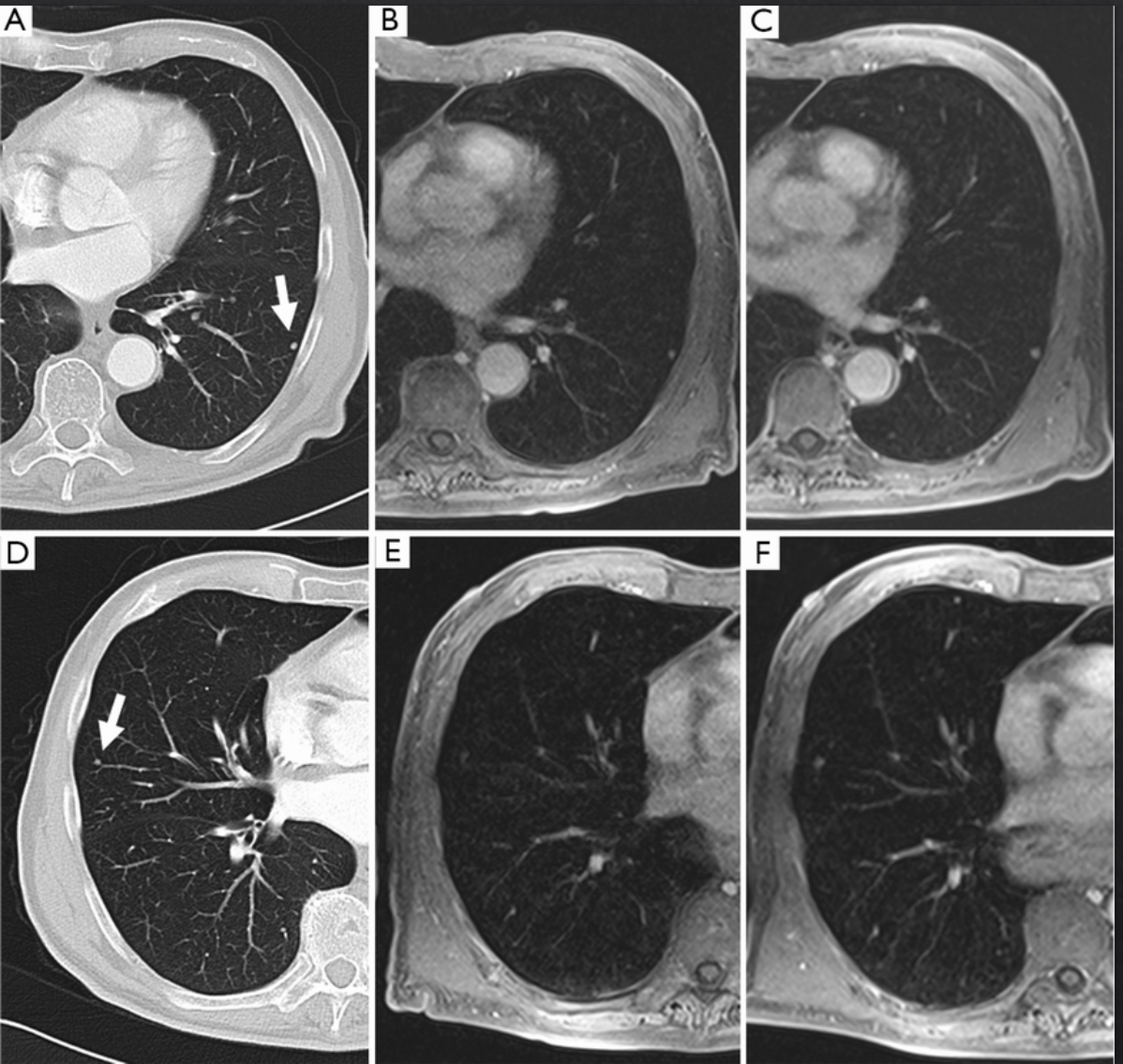
Olika MR kontraster kan leda till bättre identifiering av diskbråck

Fysiologi

- MR används flitigt inom fysiologi för att visualisera bade anatomiska och funktionella defekter för hjärtat.
- 4D (3D+tid) sekvenser kan användas som stödteknik för andra tekniker



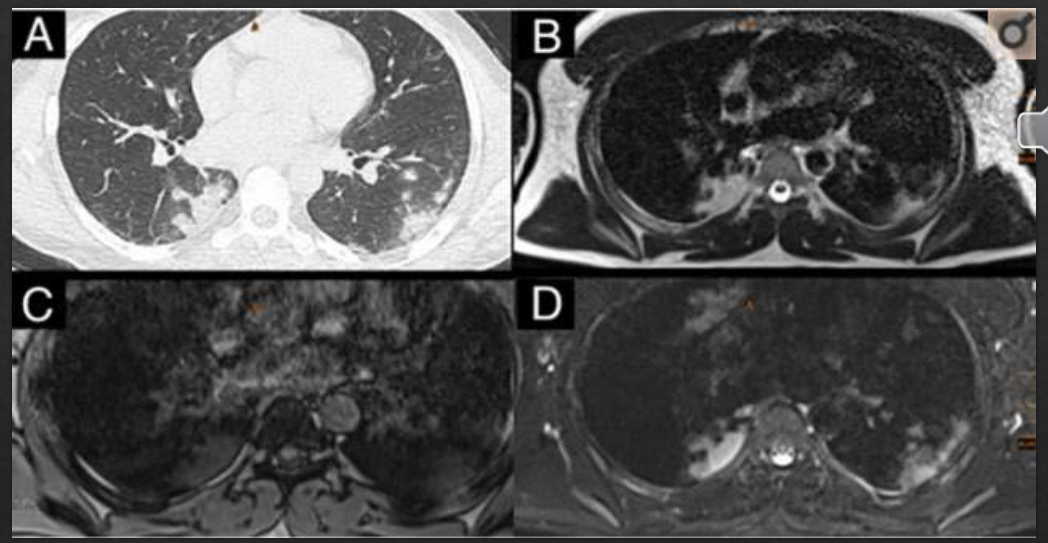
Thorax



- Oncologi
- Svår för MR – mycket luft, men nya UTE/ZTE sekvenser ger hopp för drastiska förbättringar inom området.

”Nyheter” – COVID19

MR Lunga

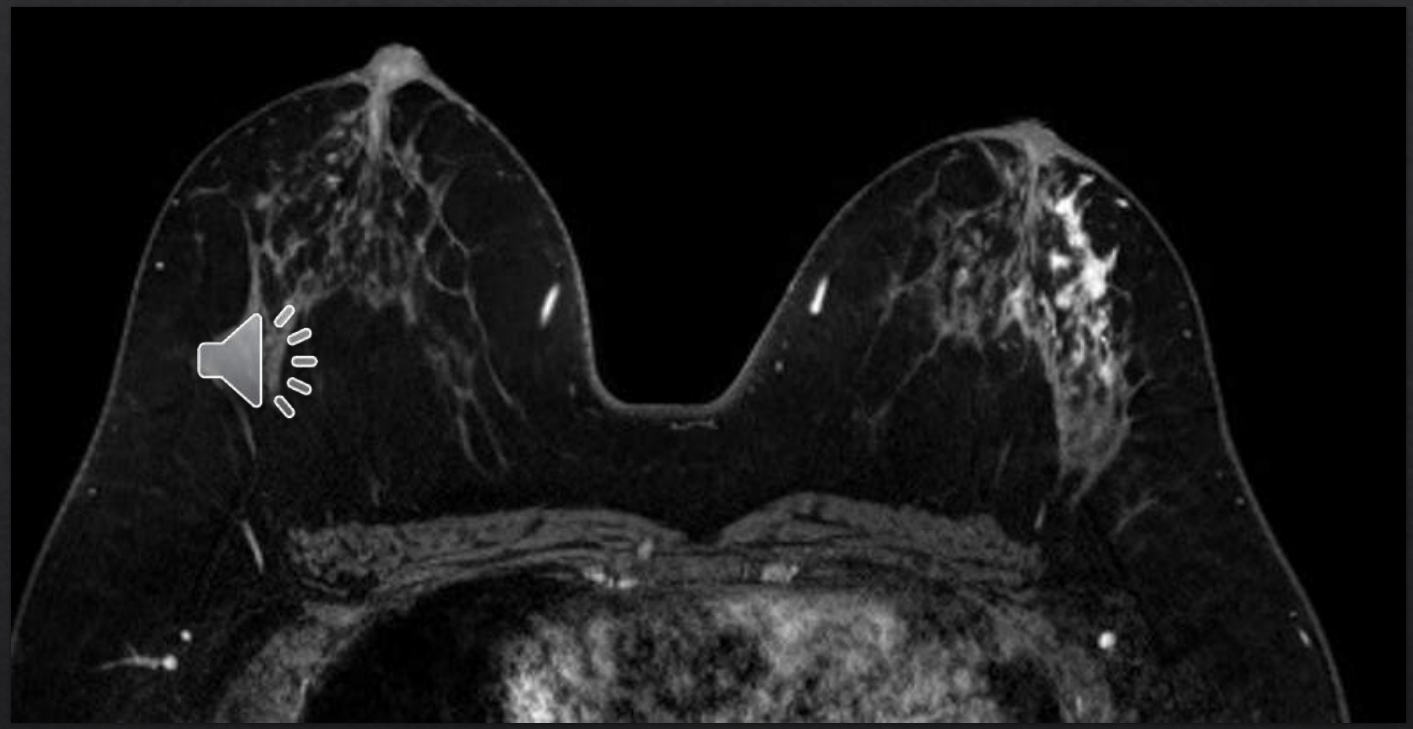


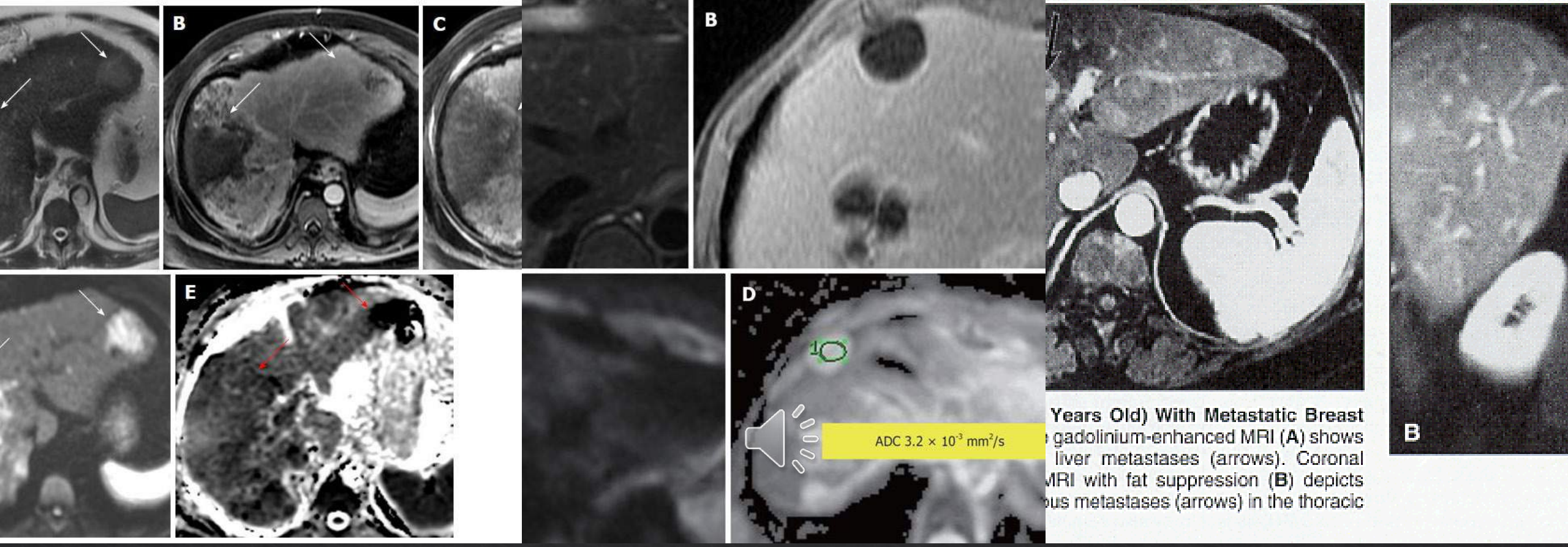
COVID-19

- Vissa COVID-19 patienter med svåra symptom visar kvarstående GGO (kristallisering av lungvävnad)
- Graden av kristallisering är direkt korrelerade med grad av symptom under, och återhämtningsförmågan.
- CT föredras pga. av enkelheten samt kostnad.

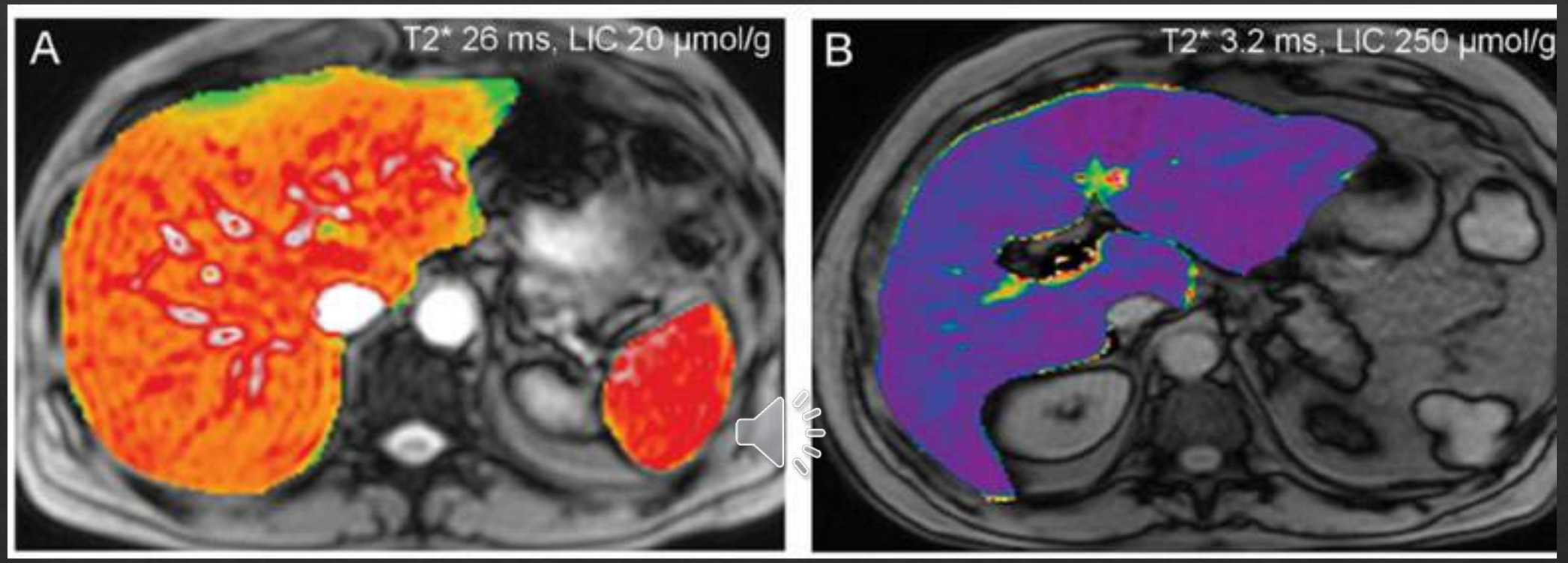
Bröst MR

- Implantat ruptur
- Onk/Pat
- Screening

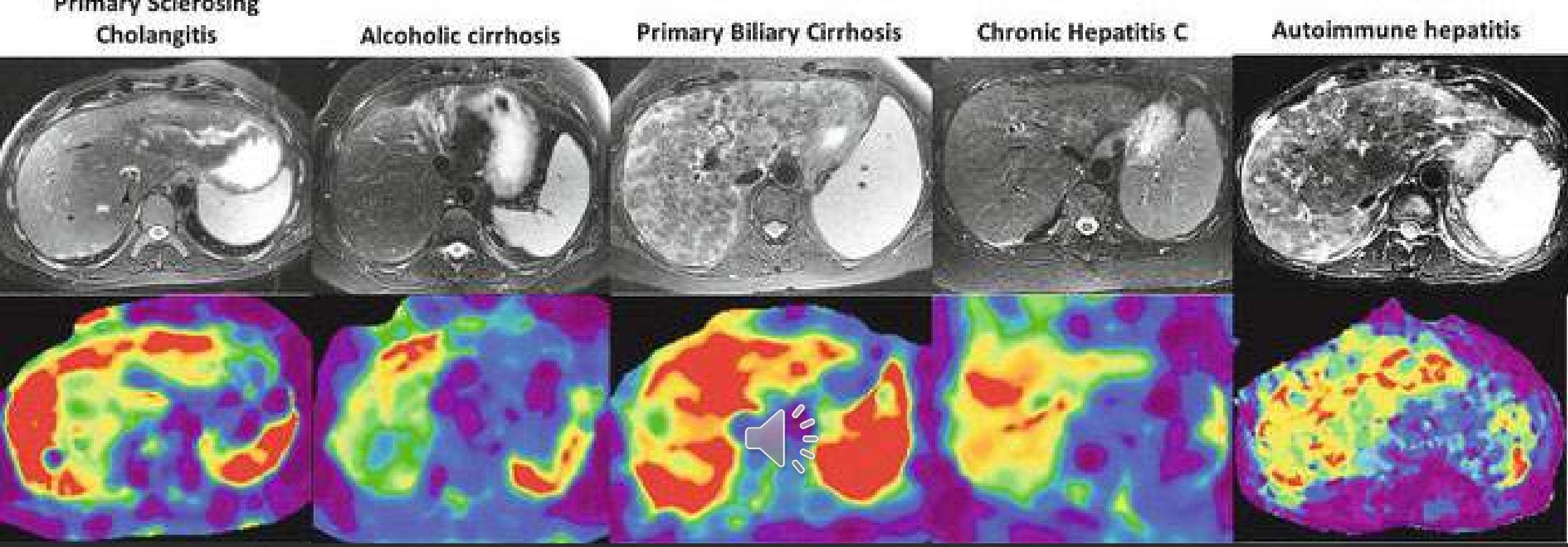




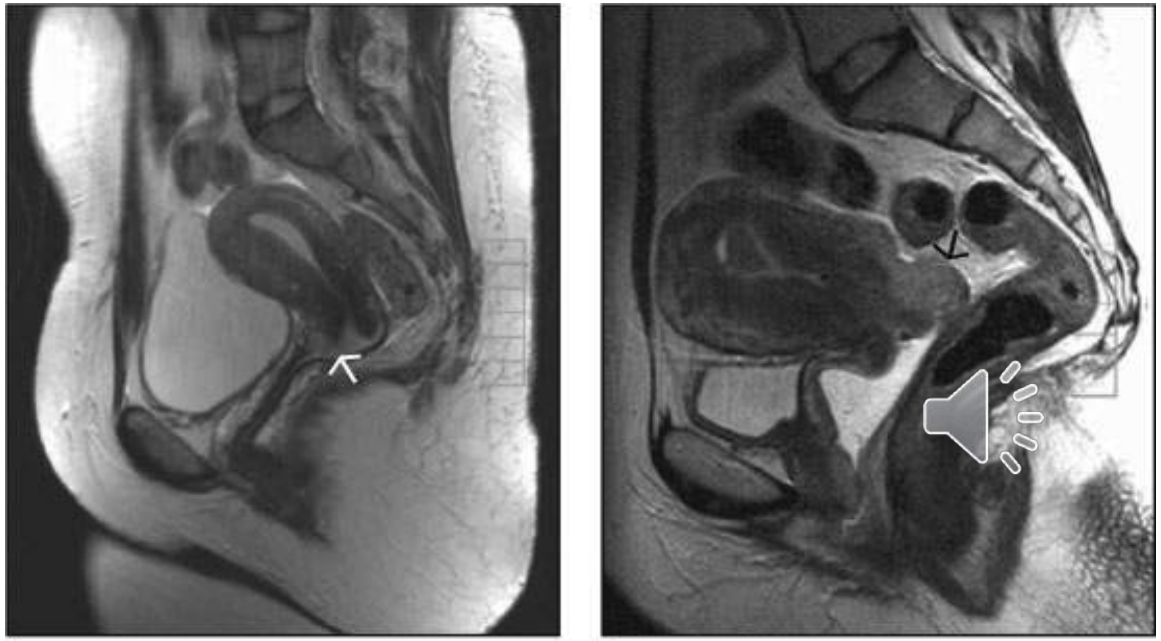
Lever, Onk/Pat



Lever, Järn



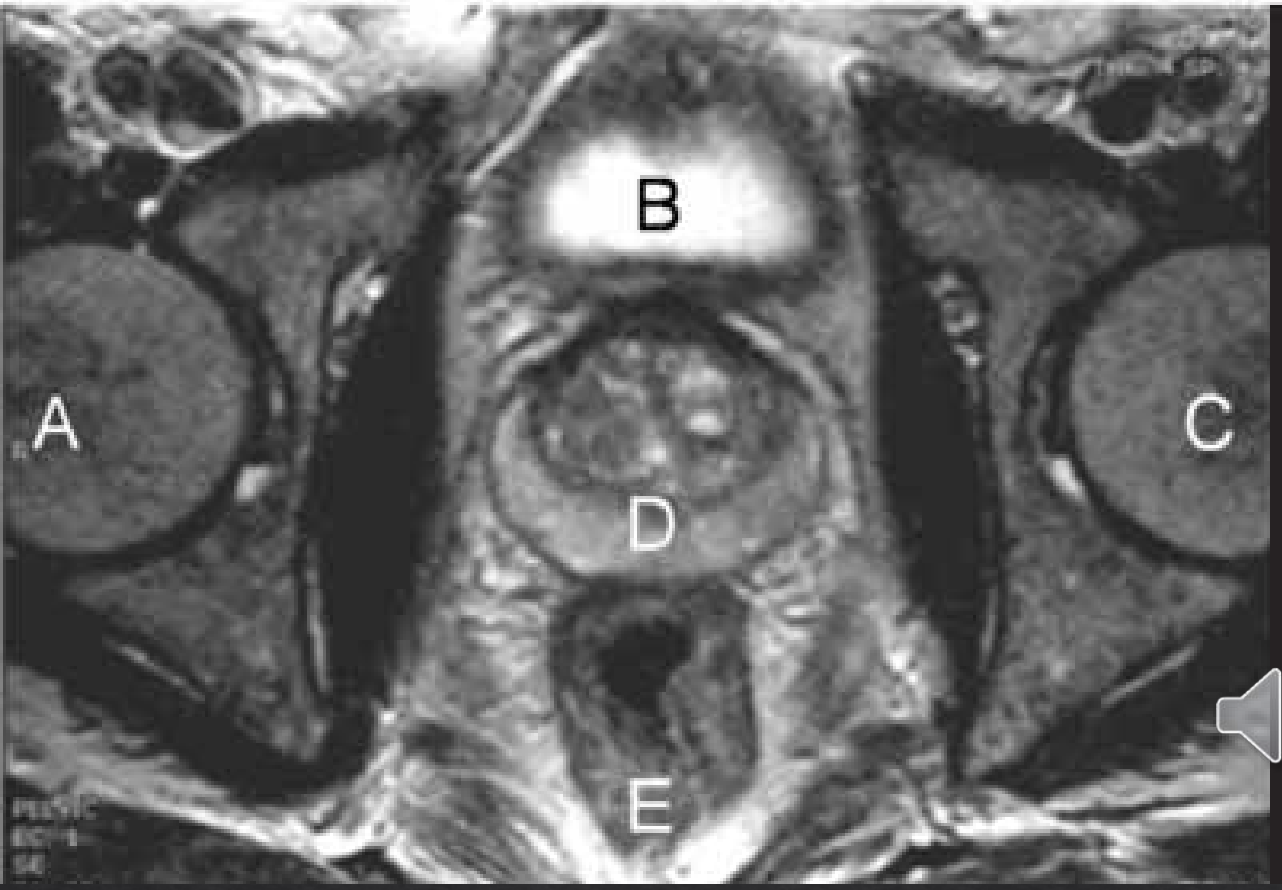
Lever, Elastografi

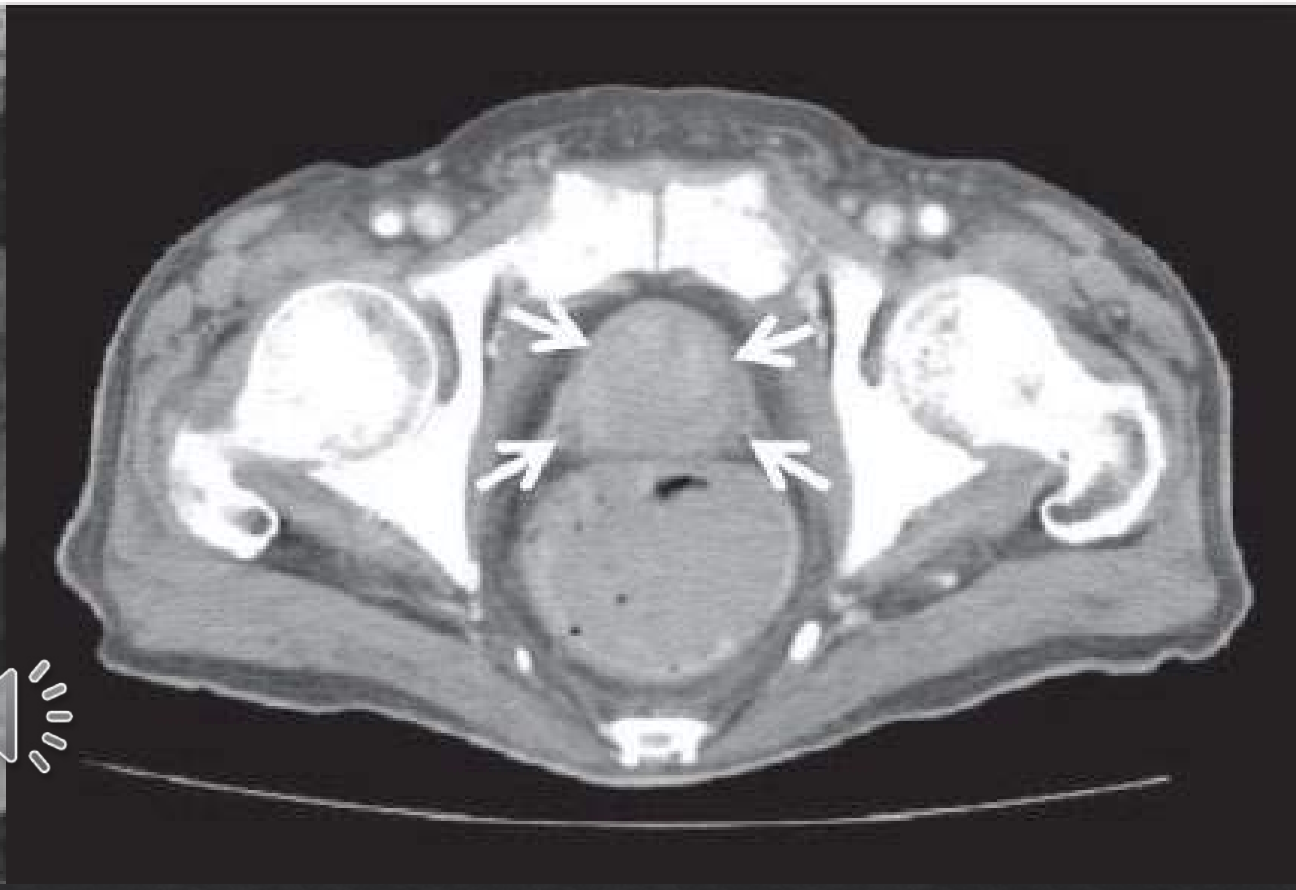


A B

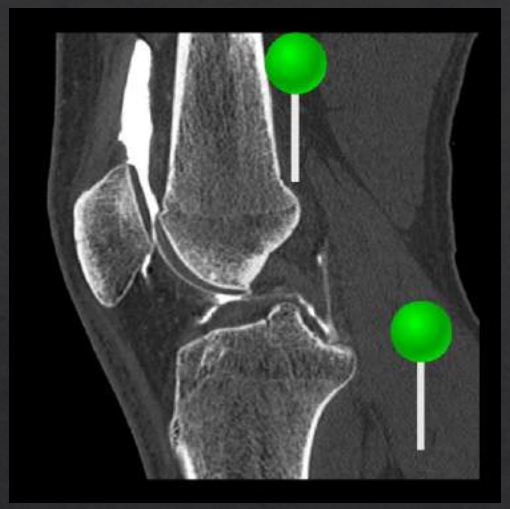
Figure 2. A: Sagittal T2-weighted TSE sequence, hyperintense uterine cervix tumor. Part of the (hypointense) cervical stroma is intact. Preserved vaginal canal (stage Ib). **B:** Sagittal T2-weighted TSE sequence, slightly hyperintense tumor in the posterior portion of the uterine cervix, extending to the upper vaginal third (stage IIa).

Buk

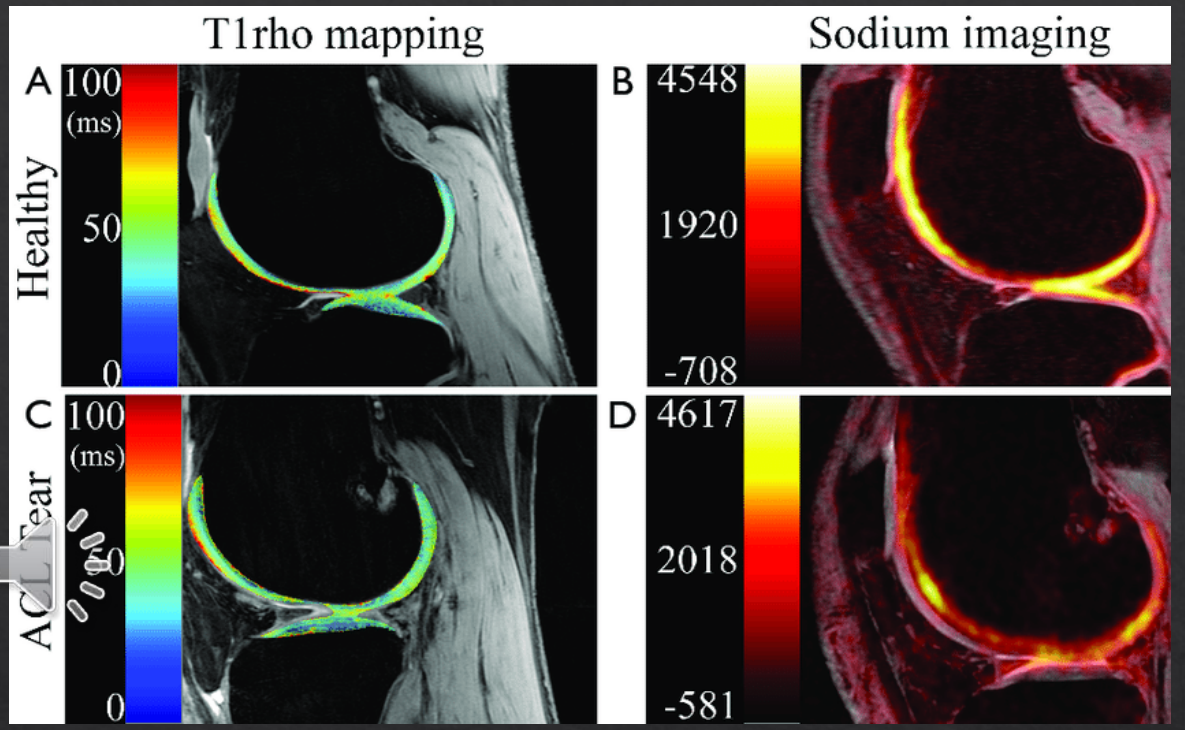




Buk, Prostata

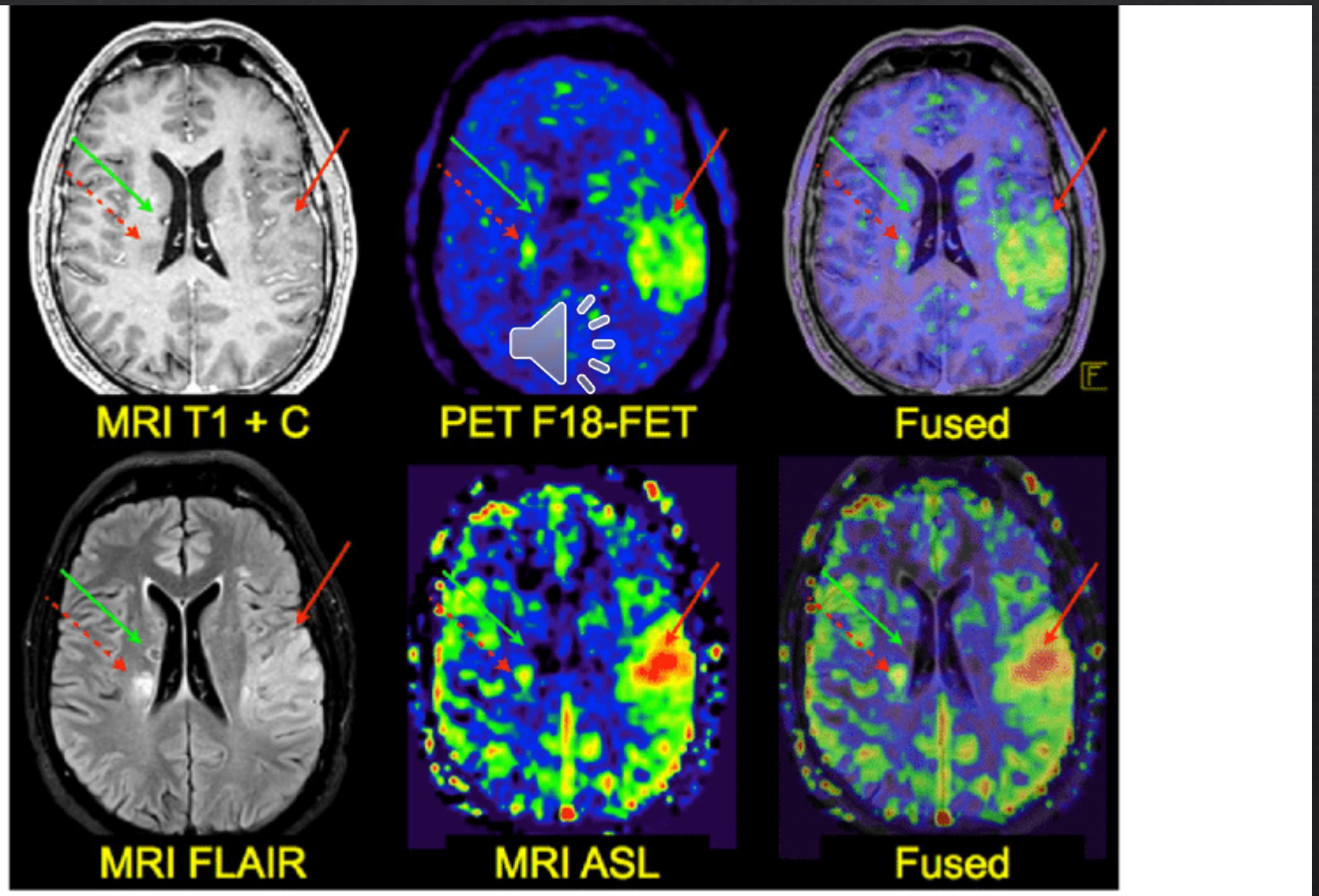





T1rho mapping
Sodium imaging
A
100
(ms)
50
0
Healthy
B
4548
1920
-708
C
100
(ms)
50
0
ACL Tear
D
4617
2018
-581

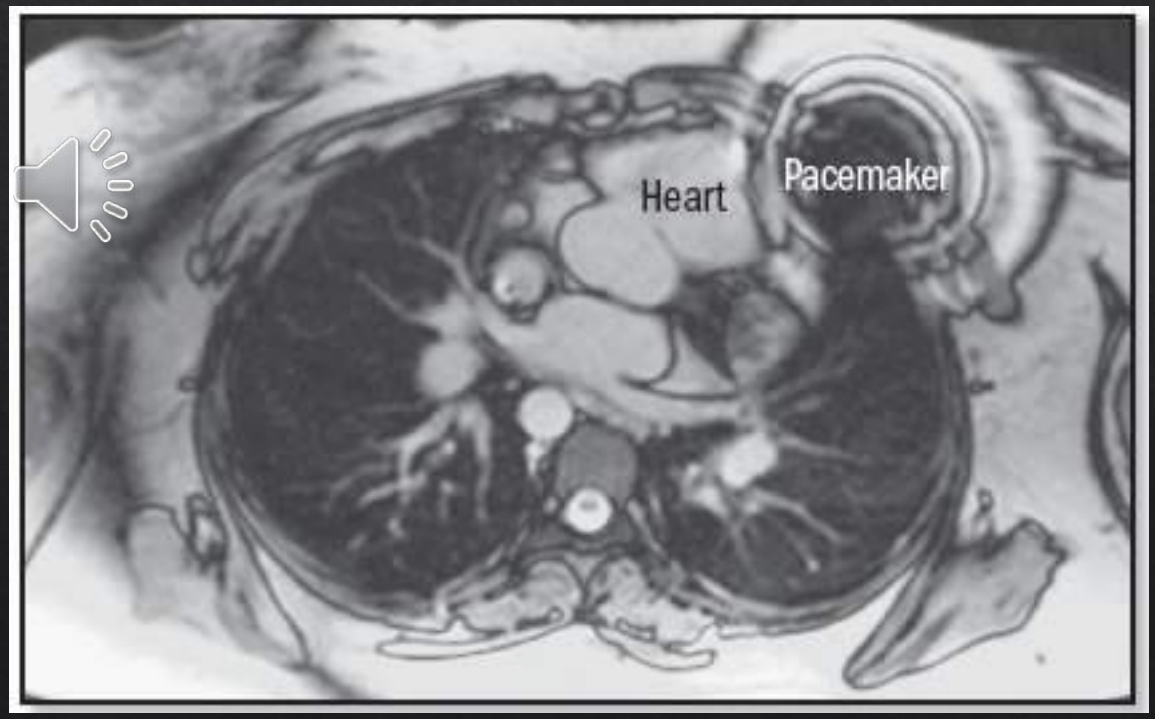
MSK

Fusion, PET/MR



Nackdelar – Metal/Elektronik

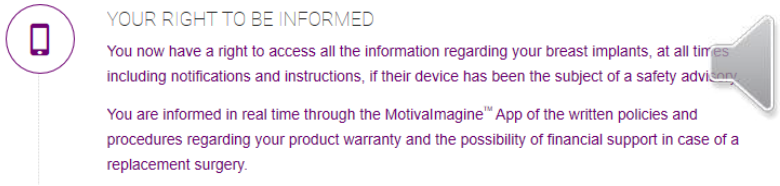


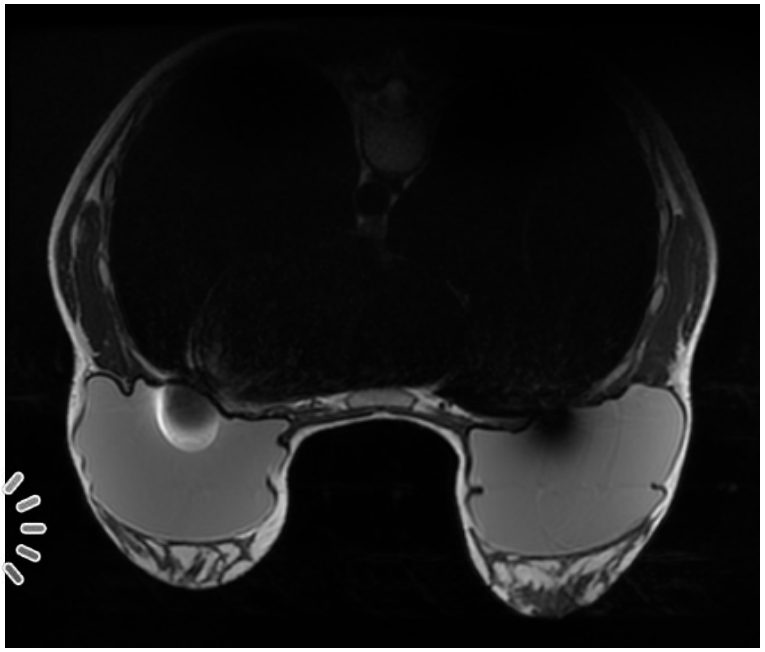


Does Electronic Identification Enablement for Silicone Gel Implants Impact Patient Safety?

Michael T Nelson[1], Kurt A Brattain[2] and Jeffrey M Williams[3]*

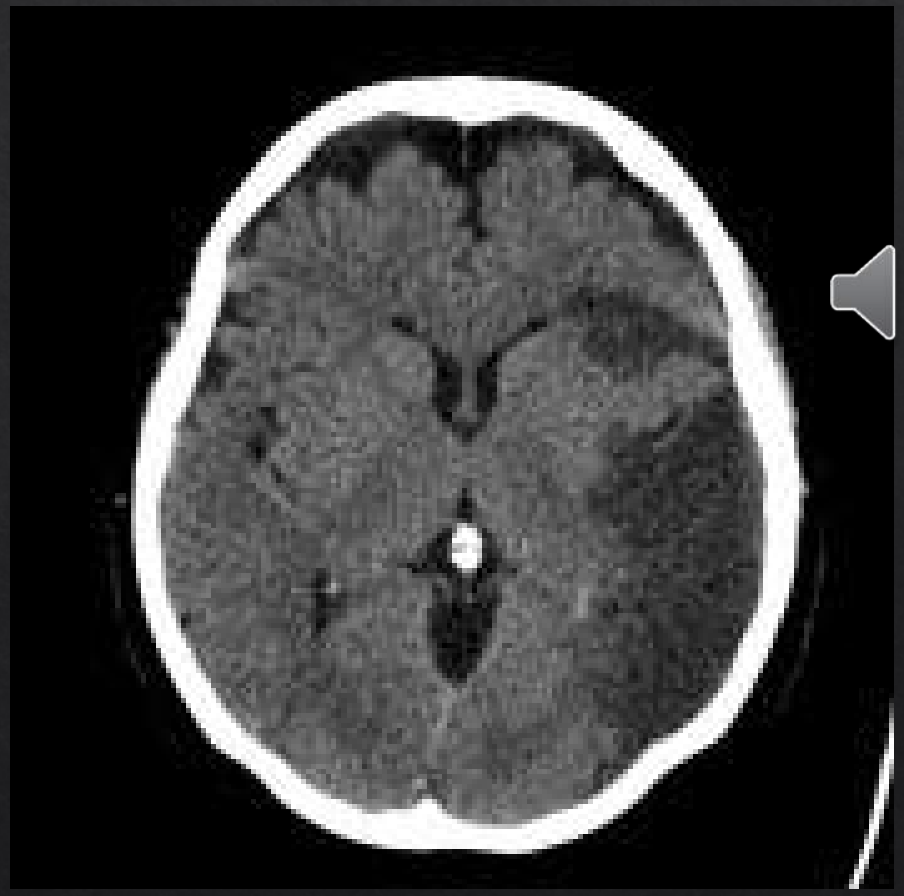
Further investigation is needed to completely address the use of RFID-M technology in silicone gel-filled breast implants and in other implantable medical devices, yet the preliminary results are promising. These early risk impact assessments RFID-M presences in the Motiva Implants® with Q Safety Inside Technology contributes to the overall safety of the patient when dual-modality imaging is utilized during cancer surveillance in high-risk patients.

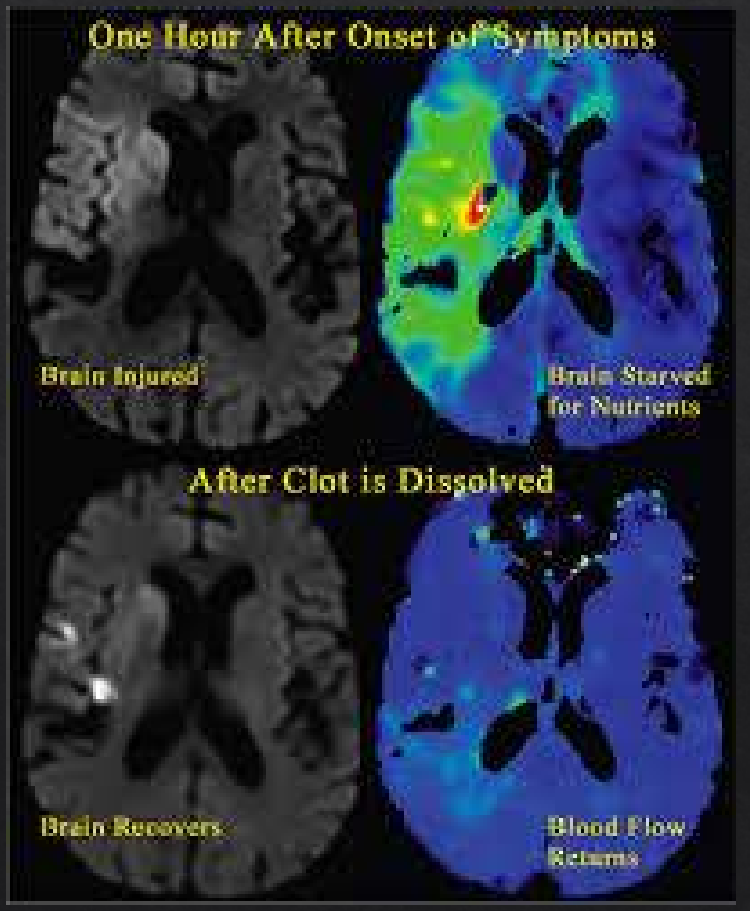




Nyhet inom plastik kirurgi

Nackdelar - Stroke









Nackdelar - klaustrofobi/ljud